

There are 4 processes in your system. Only 1 process can run at a time.

P1, arrival time = 0, burst time = 3

P2, arrival time = 1, burst time = 6

P3, arrival time = 2, burst time = 3

P4, arrival time = 2 (arrives after P3), burst time = 1

If a process arrives at time 0, requests a time slice of 1 unit, and is executed immediately, it will complete at time 1. Use FCFS for all tiebreaks except Round Robin. Answer the next 5 questions.

P ₁	0	3
P ₂	1	6
P ₃	2	3
P ₄	2	1

Question 14 [1 mark]

What is the completion time for all four processes under First Come First Serve (FCFS)?

A. P1:4, P2:10, P3:13, P4:11

B. P1:3, P2:9, P3:13, P4:10

C. P1:3, P2:12, P3:9, P4:10

D. P1:3, P2:10, P3:12, P4:13

E. P1:3, P2:9, P3:12, P4:13

$P_1 < P_2 < P_3 < P_4$

$P_1: 3; P_2: 9; P_3: 12; P_4: 13$

Question 15 [1 mark]

What is the completion time for all four processes under Shortest Job First?

A. P1:4, P2:13, P3:7, P4:5

B. P1:3, P2:9, P3:12, P4:13

C. P1:3, P2:13, P3:7, P4:4

D. P1:3, P2:10, P3:13, P4:4

E. P1:3, P2:13, P3:10, P4:4

$P_1: 3; P_4: 4; P_3: 7; P_2: 13$

Question 16 [1 mark]

What is the completion time for all four processes under Shortest Remaining Time First (a.k.a Shortest Job-to-Completion First)?

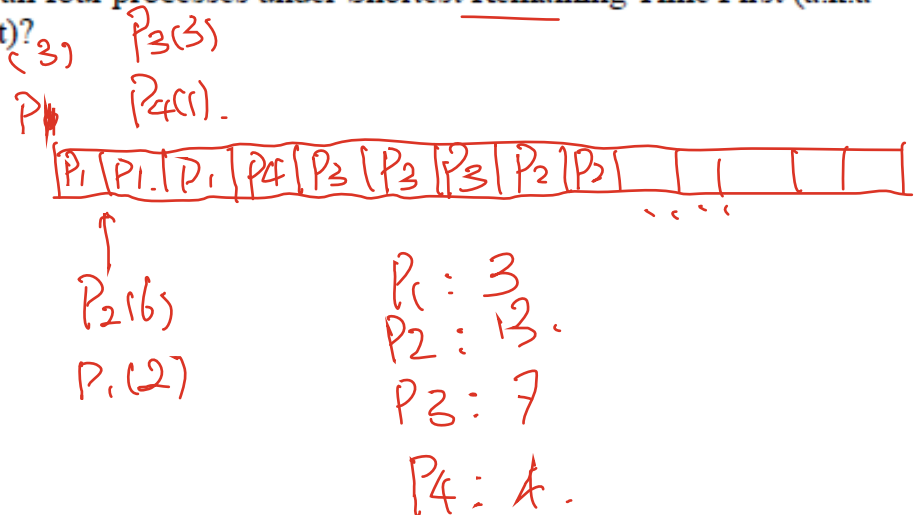
A. P1:6, P2:13, P3:9, P4:4

B. P1:3, P2:12, P3:9, P4:4

C. P1:6, P2:13, P3:7, P4:4

D. P1:3, P2:13, P3:7, P4:4

E. P1:4, P2:13, P3:7, P4:3



Question 17 [1 mark]

What is the completion time for all four processes under Round Robin with time slice of 2?

- A. P1:7, P2:13, P3:11, P4:8
- B. P1:3, P2:12, P3:13, P4:8
- C. P1:8, P2:12, P3:11, P4:7
- D. P1:8, P2:13, P3:11, P4:7
- E. P1:8, P2:11, P3:13, P4:7

Question 18 [1 mark]

After each process finishes its current CPU burst, it may later return to the ready queue and request another CPU burst of the same length as above. The time at which each process returns is unpredictable, and this may continue indefinitely. Which scheduling algorithms can guarantee that a process that is ready to run will eventually be scheduled?

- A. FCFS only
- B. Round Robin only
- C. FCFS and Round Robin only
- D. SRTF only
- E. SJF only

Question 19 [1 mark]

A company is building software for a multi-core scientific simulation. The application launches a fixed number of long-running worker processes at startup. During each phase, a worker repeatedly reuses much of the same data from recent computations. Each worker repeatedly performs similar computations throughout the run, and the problem has been divided so that the workers are expected to remain similarly busy for most of the execution. They are considering three scheduler designs:

Design A: one global ready queue shared by all CPUs

Design B: one ready queue per CPU, with no movement between CPUs

Design C: one ready queue per CPU, with frequent movement of work between CPUs

Which design is the most appropriate for this scenario?

- A. Design A or C are equally suitable
- B. Design A
- C. Design C
- D. None of the above
- E. Design B